

# Manifold Calculus of Functors and Classification of Homogeneous Functors

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# Topology

## Not a Definition

A topological space  $X$  is a set of “points” with a choice of patches of points. These patches represent the relative location between points. We call these patches “open sets”.

## Example

The set of real numbers  $\mathbb{R}$  and open intervals  $(a, b)$ , for  $a < b$ , as open sets.

For any positive  $n$ , the  $n$ -dimensional vector space  $\mathbb{R}^n$  and all the vectors in an open ball around some vector  $v_0 \in \mathbb{R}^n$  as open sets.

# Continuous maps

## Not a Definition

Given two spaces,  $X$  and  $Y$ , a continuous map  $f : X \rightarrow Y$ , is a function that sends every point  $x \in X$  to a point  $y \in Y$ , such that, if two points are “near by” in  $Y$ , then there exists two points “near by” in  $X$  that are mapped to the two points.

## Examples

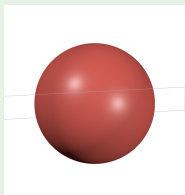
- Real continuous functions!!  $f(x) = y$
- Linear transformation over  $\mathbb{R}^n$

# Manifolds

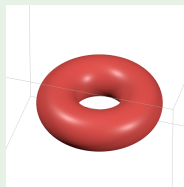
## Not a Definition

A ( $n$ -dimensional) manifold is a topological space that is “locally flat”. That is for every point  $p$  on a manifold  $M$ , there is a small patch of  $M$  containing  $p$  that looks like an open balls of  $\mathbb{R}^n$ .

## Examples



(a) 2-sphere, the vectors of  $\mathbb{R}^3$  that are length 1



(b) Torus

Figure: Some 2 dimensional manifolds

# Not a Manifold

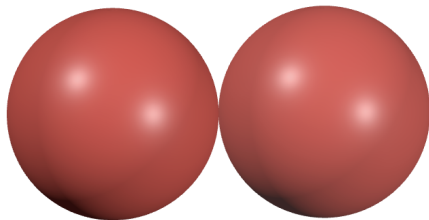


Figure: “Wedge Sum” of 2 2-spheres

Look at the part they touch, that is a point with no open neighbourhood that is flat.

# Studying Manifolds

## Question

Manifolds and continuous maps between manifolds are very difficult to study. Therefore, we would often like to capture the topological information with help of other maths (linear algebra, abstract algebra, etc.) How do we do that?

## Answer

Category Theory!!

# Category Theory

## Definition

A category  $\mathcal{C}$  is an object that “imitate a mathematical theory”.  
 $\mathcal{C}$  is made of:

- objects to study,  $obj(\mathcal{C})$  (for instance, the class of sets, where every object is a set)
- functions, or we call morphisms  $f : A \rightarrow B$ , between any two objects  $A, B \in obj(\mathcal{C})$ , such that:

# Category Theory (Cont.)

## Definition (Cont.)

- ① For every object  $A$ , there is an *identity* morphism  $id_A : A \rightarrow A$  (think about the function  $f(x) = x$  from a set back to itself)
- ② For any morphism  $f : A \rightarrow B$  and  $g : B \rightarrow C$ , we can compose them along  $B$  to create a new morphism  $g \circ f : A \rightarrow C$  (think about functions between sets, take the output of  $f$  and put it through  $g$ )
- ③ The composition is associative  $f \circ (g \circ h) = (f \circ g) \circ h$  (Does not matter here)



# Examples of Categories

## Examples

The category of sets,  $\mathbf{Set}$ , where

- objects are sets (natural numbers  $\mathbb{N}$ , real numbers  $\mathbb{R}$ , the set of mammals {elephants, whales, dogs,...}).
- the morphisms between two sets  $f : A \rightarrow B$  are functions from  $A$  to  $B$ . For instances,  $A = B = \mathbb{R}$ , and  $f(x) = x^2$ .

The category of topological spaces,  $\mathbf{Top}$

- objects are spaces (spheres, tori,  $\mathbb{R}^n$ ).
- the morphisms are continuous maps.

The category of vector spaces,  $\mathbf{Vec}_{\mathbb{F}}$

- objects are vector spaces
- the morphisms are linear transformation.

# Categories that are not Maths Theory

## Example

For instance, given the finite set  $K := \{x, y, z\}$ , its power set  $P(K) := \{\text{subsets of } K\}$  is also a category, where the objects are the subset of  $K$ , and morphisms are inclusions of subsets.

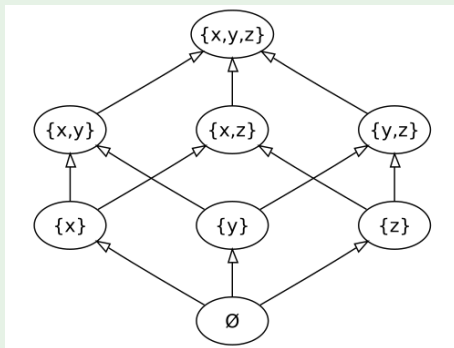


Figure: The category of the powersets of the set  $\{x, y, z\}$

# Category of Open sets

## Manifold as Category

The category of open sets of a manifold  $M$ ,  $\mathcal{O}(M)$  is made of

- objects are the open sets of  $M$
- for any two open sets  $U, V \subseteq M$ , there exists one and only one  $U \rightarrow V$  if  $V$  contains  $U$ , (empty otherwise)

This category captures the information of  $M$  entirely.

# From Category to Category

## Question

Given two category  $\mathcal{C}$  and  $\mathcal{D}$ , where we want to study  $\mathcal{C}$ , and we know how to study  $\mathcal{D}$ . How do we use methods in  $\mathcal{D}$  to study  $\mathcal{C}$ ?

## Answer

Functors!!

# Covariant Functors

## Not a Definition

A covariant functor  $F : \mathcal{C} \rightarrow \mathcal{D}$  takes an object  $c$  of  $\mathcal{C}$  to an object  $F(c)$  of  $\mathcal{D}$ , and takes morphism  $f : c \rightarrow c'$  of  $\mathcal{C}$  to  $F(f) : F(c) \rightarrow F(c')$  of  $\mathcal{D}$  (plus some requirements)

## Example

- The “forgetful” functor  $\text{fgt} : \text{Top} \rightarrow \text{Set}$  that takes a space  $X$  to its set of points, and continuous maps to their underlying functions.
- There are some functors called “homology”,  $H : \text{Top} \rightarrow \text{Vec}$  that take hard-to-deal-with topological information to easy-to-deal-with linear algebra.

# Contravariant Functors

## Not a Definition

Essentially, the same as covariant functors, but this time the direction of morphisms are changed, so we have  $f : c \rightarrow c'$  are take to  $F(f) : F(c') \rightarrow F(c)$ . These are often more interesting to study. We denote such functor as  $F : \mathcal{C}^{op} \rightarrow \mathcal{D}$ .

## Example

There are some functors called “cohomology” takes spaces to not just vector spaces, but to something called “algebra” (essentially, vector spaces with multiplication of any two vectors)

# Calculus of Functors

For any category  $\mathcal{C}$ , a contravariant functor  $F : \mathcal{O}(M)^{op} \rightarrow \mathcal{C}$  can be complicated. Therefore, just as how we approximate smooth functions with its Taylor series

$$f(x) \approx \sum_{i=0}^k \frac{f^{(i)}(0)}{i!} x^i$$

We might want to approximate  $F$  with a “Taylor series”.

# Manifold Calculus of Functors

Weiss in [6] developed the Manifold Calculus of Functors when  $\mathcal{C} = \text{Top}$ , and the concepts were generalized later to an arbitrary category in [5]

## Approximations

For an functor  $F : \mathcal{O}(M)^{op} \rightarrow \mathcal{C}$ , there is a sequence of functors

$$\mathcal{T}_k F : \mathcal{O}(M)^{op} \rightarrow \mathcal{C}$$

whose value on an open set  $U$  is obtained by "gluing" the information of  $F$  over the disjoint union of disks of at most  $k$  components contained in  $U$ .

The functor  $\mathcal{T}_k F$  would recover parts of the information of  $F$ , as  $k$  increases, more information is recovered.



# Homogeneous Polynomials

## Homogeneous Polynomials

The polynomial functions in the form  $g(x) = ax^k$  are called homogeneous. They have some nice properties.

- The  $k$ th Taylor approximation of  $g(x)$  is itself
- The  $k - 1$ th Taylor approximation is 0

These types of functions are the building blocks of functions, and they are relatively easy to work with in Calculus.

Similarly, we would like to work with “homogeneous functors” in Manifold Calculus.

# Homogeneous Functors

## Not a Definition

Let  $k > 0$ , a functor  $G : \mathcal{O}(M)^{op} \rightarrow \mathcal{C}$  is called a *homogeneous functor of degree  $k$*  if for open set  $U$  of  $M$ ,  $G(U)$  is obtained as "gluing" the information of  $G$  over disjoint union of exactly  $k$  disks contained in  $U$ .

The functor  $G$  is a polynomial functors of degree  $\leq k$  such that  $\mathcal{T}_{k-1}G$  is trivial, because we do not need the informations on  $k - 1$  and less components of disks.

# What are the possible Homogeneous Functors?

## Question

The definition of homogeneous functors is very technical. Is there a simpler characterization of homogeneous functors over a given manifold?

# Weiss's Classification of Homogeneous Functors

Weiss in [6] provided a characterization of Homogeneous functors when  $\mathcal{C} = \text{Top}$ .

# Fibrations

## Not a Definition

Given topological spaces  $B$  and  $F$ , a fibration over  $B$  with fiber  $F$  is a continuous map  $\pi : M \rightarrow B$ , such that, for every point on  $x \in B$ , there is an open set  $U \subseteq B$  containing  $x$ , such that all the points of  $m \in M$  such that  $\pi(m) \in U$ , looks like  $F \times U$ .

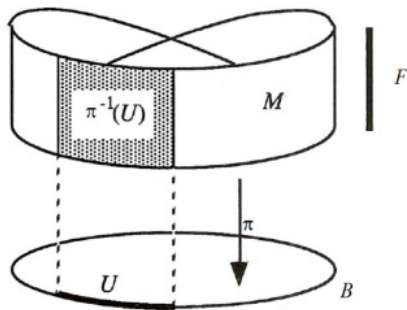


Figure: Picture From [1]

# Disks and Points of a Manifold

## Not a Definition

The space of unordered configuration of  $k$  points of a manifold  $M$ ,  $F_k(M)$ , is the space where every point  $S \in F_k(M)$  corresponds to  $k$  distinct points on  $S \subset M$ .

We see that every point  $S$  of  $F_k(M)$  determines an open set  $U_S$  on  $M$  that looks like a disjoint union of  $k$  disks.

# Homogeneous Functors in Top

As it turns out, Weiss showed in [6, Part 8], that every homogeneous functor of degree  $k$ ,  $F$ , such that the value of  $F$  on the disjoint union of  $k$  disks is  $A$  arise from a fibration  $p : Z \rightarrow F_k(M)$  with fiber  $A$ . We will refer to these homogeneous functors as type  $kA$ .

# Homogeneous Functors in Top (Cont.)

Moreover, by [3], the class of fibration over a space  $B$  with fiber  $F$  is classified by the equivalent types of continuous maps from  $B$  to  $B\text{haut}(F)$ . Here  $B\text{haut}(F)$  is called the classifying space of  $F$ -fibration.

## Theorem

[6] For any manifold  $M$ , and any topological space  $A$ , we have

$$\{\text{Homog. of Type } kA / \sim\} \cong \text{Map}(F_k(M), B\text{haut}(A)) / \sim$$



# Tsopmén  and Stanley's Classification of Homogeneous Functors

Now suppose we want to classify the homogeneous functors in any  $\mathcal{C}$ . It is impossible to replicate the method used by Weiss.

Tsopmén  and Stanley in [4] provided a different strategy to classify the homogeneous functors. In which, for every object  $A \in \mathcal{C}$ , they constructed a topological space  $\hat{A}$ .

# Manifold as Polytopes

Any (smooth) manifold  $M$  can be recovered by gluing numbers of polytopes in its dimension. We call this a triangulation.

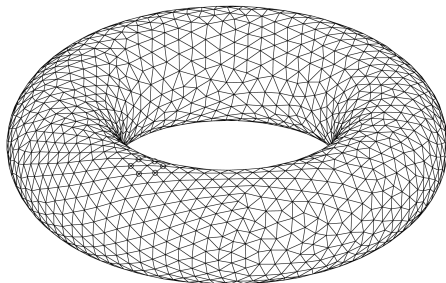


Figure: A triangulation of the Torus

# Diagrams in $\mathcal{C}$

The space  $F_k(M)$  is also an manifold, and a homogeneous functor of degree  $k$  over  $M$  is equivalent to a homogeneous functor of degree 1 over  $F_k(M)$ . Therefore, it suffices to look at the homogeneous functors of degree 1.

Given a triangulation of  $M$ , a homogeneous functor is determined by its value over each of the polytopes

The space  $\hat{A}$  is defined to parametrize all the diagrams in  $\mathcal{C}$  that comes from a polytope with values on the object equivalent to  $A$ .

# Homogeneous Functors in Any $\mathcal{C}$

Then there is the following

## Theorem

[4, Theorem 1.1] For any manifold  $M$  and any object  $A \in \mathcal{C}$

$$\{\text{Homog. of Type } kA / \sim\} \cong \text{Map}(F_k(M), \hat{A}) / \sim$$

# How are Two Classification Connected

The space  $B\text{haut}(F)$  classifies  $F$ -fibration, but its construction is irrelevant of the concept of fibration. In other words, for any category  $\mathcal{C}$  and any object  $A \in \mathcal{C}$ , we are able to construct  $B\text{haut}(A)$ .

## A Conjecture

Tsopmén  and Stanley in [4, Conjecture 8.3] conjectures that their space  $\hat{A}$  equivalent to the space  $B\text{haut}(A)$ .

If this is true, then we are able to connect the classification of Tsopmén  and Stanley to that of Weiss, seeing as a generalization of the result.

# Our Result

As it turns out, this is indeed true. In our paper, we proved that

## Theorem

[2, Theorem 1.1] For any object  $A \in \mathcal{C}$ , there is equivalence

$$\hat{A} \simeq B\mathrm{haut}(A)$$

The proof constructed a zigzag of equivalence

$$\hat{A} \xleftarrow{\simeq} \dots \xrightarrow{\simeq} \dots \xleftarrow{\simeq} B\mathrm{haut}(A)$$

that removes redundant information in each step.

# Why Do We Care?

First of all,  $\hat{A}$  is hard to work with, but  $B\text{haut}(A)$  is cool!!

The space  $\hat{A}$  comes from an extremely technical construction, which requires one to have a very good understanding of the category  $\mathcal{C}$ . Whereas the space  $B\text{haut}(A)$  comes from a relatively easy construction.

## Example (Computation)

Some computation that you can do at home!! Let  $A = \mathbb{C} \in \text{Vec}_{\mathbb{C}}$  be the 1-dimensional complex vector space, one can easily (?) compute the cohomology of  $B\text{haut}(\mathbb{C})$ .

# Connection to Physics

For some objects  $A$ ,  $B_{\text{haut}}(A)$  is called the classifying space of principal  $G$ -bundle, where  $G$  is a type of group called Lie group. Lie groups are the heart of modern physics.

## Example (Gauge Theory)

In physics, one cares about something called the connections of principal  $G$ -bundle over  $M$ . As it turns out, the equivalent types of connections over  $M$  is equivalent to the types of homogeneous functors of  $1A$  over  $M$ .

## Example (TQFT)

The connections of principal  $G$ -bundle determines what we call a TQFT.



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